VyapaarSetu

AI-Powered MSE Agent Mapping Tool for ONDC TEAM Initiative

Performance Indicators & Benchmarking Report

Problem Statement:	AI-Powered MSE Agent Mapping Tool (PS2)
Ministry:	Ministry of Micro, Small and Medium Enterprises (MSME)
Challenge:	IndiaAI Innovation Challenge 2026
Document Version:	1.0
Date:	February 2026

1. Executive Summary

VyapaarSetu is an AI-powered MSE Agent Mapping Tool designed to automate ONDC onboarding under the TEAM initiative (Rs. 277 Cr, FY2024-27). This document presents comprehensive performance benchmarks demonstrating how VyapaarSetu improves upon existing manual processes and open-source models through innovative AI/ML integration across three core capabilities: multilingual voice-enabled registration, AI-driven product cataloguing, and intelligent MSE-to-SNP matching.

Key Achievement	Metric	Result
Voice Response Latency	End-to-end response time	<3 seconds average
Product Classification	ONDC taxonomy accuracy	88-92% (vs 55-60% baseline)
Attribute Extraction	Voice-to-structured data	95% accuracy
SNP Matching Relevance	Top-3 precision	90% relevant matches
RAG Retrieval	ONDC knowledge precision	90% top-3 retrieval
Onboarding Time	Voice vs manual forms	5 min vs 30-60 min

2. Detailed Performance Metrics

2.1 ONDC Product Classification Accuracy

Metric:

Classification accuracy across 10 ONDC retail categories (RET10-RET19): Grocery, Food & Beverages, Fashion, Beauty, Electronics, Appliances, Home & Decor, Toys, Health & Wellness, Pharma

Methodology:

LLM-based classification with ONDC taxonomy context injection via Retrieval-Augmented Generation (RAG). Tested on 100+ product descriptions across categories with known ground-truth ONDC RET codes.

Baseline Performance:

Generic LLMs without ONDC domain context achieve approximately 55-60% accuracy on ONDC retail taxonomy classification due to overlapping categories (e.g., herbal products could be RET13, RET18, or RET10).

VyapaarSetu Result:

88-92% classification accuracy with RAG-injected ONDC taxonomy specifications

Improvement:

+30-35% improvement over baseline through domain-specific knowledge grounding

False Positive Rate:

<8% (products assigned to wrong but adjacent category)

False Negative Rate:

<5% (products left unclassified)

2.2 Voice Agent Response Latency

Metric:

End-to-end response time from user speech input to agent speech output

Pipeline Components:

Speech-to-Text (Sarvam AI IndicTrans2) + LLM Processing (DeepSeek, single-call architecture) + Language Detection (Sarvam AI LID) + Text-to-Speech (Sarvam AI)

Methodology:

Measured across complete voice pipeline with real-time Hindi/Hinglish conversation scenarios for product cataloguing

Component Breakdown:

STT: ~1 second | LLM: ~0.8 second | LID: ~0.2 second | TTS: ~1 second

VyapaarSetu Result:

Under 3 seconds average end-to-end latency

Improvement:

60-70% faster than ReAct agent with tools (5-10 seconds) due to single-call Voice Agent architecture

2.3 Voice Field Extraction Accuracy

Metric:
Accuracy of structured field extraction from natural voice conversations in Hindi/Hinglish

Fields Extracted:
product_name, mrp, material, moq, hsn_code, country_of_origin (6 core fields per product)

Methodology:
Voice Agent extracts fields using JSON [FIELDS] tags in structured output format. Tested across 50+ voice conversations with varied speaking styles, mixed Hindi-English input, and numeric values.

VyapaarSetu Result:
95% extraction accuracy for clearly spoken inputs

False Positive Rate:
<5% (incorrect field values extracted — e.g., wrong MRP)

False Negative Rate:
<10% (missed fields requiring re-asking via WhatsApp follow-up)

Key Insight:
Multi-field extraction in a single utterance (e.g., "Peetal ka diya banata hoon, 150 rupay, 10 piece minimum") achieves 90%+ accuracy due to LLM's contextual understanding.

2.4 RAG Retrieval Relevance

Metric:
Retrieval precision for ONDC taxonomy and TEAM scheme knowledge queries

Knowledge Base:
ONDC retail taxonomy (RET10-RET19 with mandatory/optional attributes), TEAM scheme guidelines, SNP capability data, E-Commerce Rules 2020 compliance requirements, Legal Metrology Act product labelling rules

Vector Database:
Qdrant with Sentence-Transformers (all-MiniLM-L6-v2) embeddings, hybrid search (vector + BM25)

VyapaarSetu Result:
90% top-3 retrieval precision for ONDC category attribute queries

Improvement:
Eliminates hallucination of ONDC compliance requirements by grounding in actual taxonomy specifications

2.5 Catalogue Compliance Validation

Metric:
Accuracy of compliance scoring against E-Commerce Rules 2020 and ONDC mandatory field requirements

Methodology:
validate_catalogue tool checks extracted attributes against category-specific mandatory field schemas. Compliance score = (present mandatory fields / total mandatory fields) × 100

VyapaarSetu Result:
100% accurate compliance scoring (deterministic rule-based validation)

False Positive Rate:

0% — product never marked "ready" with missing mandatory fields

2.6 SNP Matching Relevance

Metric:

Relevance of top-3 SNP recommendations based on MSE profile, product categories, geography, and transaction type

Methodology:

Weighted scoring algorithm: Category Overlap (45%) + Geography Coverage (25%) + Transaction Type Alignment (20%) + Onboarding Capacity (10%). Evaluated against 6 real ONDC SNPs (GlobalLinker, Plotch, SellerApp, eSamudaay, DotPe, MyStore).

VyapaarSetu Result:

90% relevant top-3 recommendations (at least 2 of 3 SNPs are genuinely suitable)

Improvement:

Eliminates manual trial-and-error SNP selection by MSEs who lack knowledge of ONDC ecosystem

2.7 Identity Verification Accuracy

Metric:

Correct match/reject of MSE's registered mobile number during voice verification

Methodology:

Voice Agent compares spoken number against database record injected in system prompt. MSE profile (name, mobile, business) loaded from PostgreSQL and injected into voice prompt context.

VyapaarSetu Result:

100% accuracy for exact matches (last 10 digits comparison)

Partial Match Handling:

Intelligent handling of 10-digit variants, country code prefixes, and format variations (with/without +91)

2.8 Multi-Channel Consistency

Metric:

Response consistency for identical product queries across Voice, WhatsApp, and Web channels

Methodology:

Identical product descriptions submitted via all three channels, classification results and extracted attributes compared for consistency

VyapaarSetu Result:

100% consistent ONDC classification and attribute extraction across all channels (unified AI pipeline)

Architecture:

Shared tool layer (classify_product, extract_attributes, validate_catalogue) ensures identical AI processing regardless of input channel. Only agent prompts differ per channel for UX optimization.

3. Improvement Over Existing Approaches

3.1 Comparison with Manual TEAM Onboarding Process

Parameter	Current Manual Process	VyapaarSetu AI
Registration Time	30-60 minutes (complex forms)	5 minutes (voice-enabled)
Product Cataloguing	Manual data entry, high error rate	AI auto-classification, 88-92% accuracy
ONDC Taxonomy Mapping	Requires trained operators	Automated LLM + RAG pipeline
Compliance Validation	Manual checklist review	Real-time automated scoring
SNP Selection	Random/geographic assignment	AI-scored matching (4 weighted factors)
Language Support	English forms only	Hindi, English, Hinglish (voice + chat)
Channel Access	Web portal only	Voice + WhatsApp + Web
Follow-up for Missing Info	Phone calls by operators	Automated WhatsApp follow-up

3.2 Comparison with Open-Source LLM Baselines

Capability	Generic LLM (No Domain Context)	VyapaarSetu (RAG + Tools)
ONDC Category Classification	55-60% accuracy	88-92% accuracy
Attribute Extraction	60-70% (misses domain-specific fields)	95% (category-aware extraction)
Compliance Knowledge	Hallucinated requirements	Grounded in actual ONDC specs
SNP Awareness	No knowledge of ONDC SNPs	6 real SNPs with capability data
Voice Response Time	5-10 sec (multi-turn ReAct)	<3 sec (single-call Voice Agent)

4. Monitoring & Enhancement Strategy

4.1 Classification Drift Monitoring

Product misclassifications are logged with user corrections. Feedback loop enables continuous improvement of few-shot examples and category disambiguation rules. Monthly review of classification accuracy across all ONDC RET categories.

4.2 Compliance Rule Updates

Validation rules are codified from E-Commerce Rules 2020 and Legal Metrology Act. When ONDC updates mandatory attribute schemas, the validate_catalogue tool's rule engine is updated without requiring model retraining.

4.3 Voice Pipeline Quality

STT accuracy monitored via user correction patterns. If users frequently repeat information, prompts are refined for clarity. TTS language detection (LID) accuracy tracked to ensure correct Hindi/English voice selection.

4.4 SNP Data Freshness

SNP capability data (categories, geography, onboarding speed) refreshed quarterly from ONDC Network Participants directory. New SNPs added to matching algorithm as they join the ONDC network.

5. Methodology Summary

Indicator	Methodology	Test Size	Result
Classification Accuracy	Ground-truth RET code comparison	100+ products	88-92%
Field Extraction	Structured output validation	50+ conversations	95%
Voice Latency	End-to-end pipeline timing	Real-time sessions	<3 sec
SNP Relevance	Expert review of top-3	30+ MSE profiles	90%
RAG Retrieval	Top-3 precision @ K	50+ queries	90%
Compliance Scoring	Deterministic rule check	All products	100%
Identity Verification	DB record comparison	All sessions	100%
Multi-Channel Consistency	Cross-channel comparison	20+ queries	100%